Experiment 5

Aim

To study software estimation techniques (LOC, Function Points, Productivity, COCOMO) to estimate the effort, time, and cost.

Case Study

Online Food Delivery System – customers order food from restaurants, delivery partners deliver it, and payment, maps and SMS/email services are integrated.

Objective

1. To understand the role of software metrics in project estimation.
2. To apply LOC, Function Point, Productivity, and COCOMO techniques for effort and time calculation.
3. To estimate cost and resources required for a software project.

Step 1: Identify Metrics

Decide which metric to use for estimation:

- **LOC (Lines of Code)** – derived from FP using the assumed language ratio (Java ≈ 53 LOC/FP).
- **Function Points (FP)** – based on inputs, outputs, files, inquiries and interfaces.
- **COCOMO Model** – uses KLOC (size) to get effort and time; salary per month gives cost.

Step 2: Collect Basic Data

For the Online Food Delivery System:

- **External Inputs (EI):** Register, Login, Add to Cart, Place Order, Add/Update Menu Item, Assign Delivery Partner, Submit Rating & Review → **7**
- **External Outputs (EO):** Order Invoice/Receipt, Order Confirmation Notification, Restaurant Sales Report → **3**
- **External Inquiries (EQ):** Track Order Status, Search Restaurants/Dishes, View Order History → **3**
- **Internal Logical Files (ILF):** Users DB, Restaurants DB, Menu Items DB, Orders DB, Delivery Partners DB → **5**
- **External Interface Files (EIF):** Payment Gateway, Maps/GPS API, Email/SMS API → **3**

Step 3: Calculate Size (FP / LOC)

Use average weights (if detailed complexity is not rated): **EI = 4, EO = 5, EQ = 4, ILF = 10, EIF = 7**.

Component	Count	Weight	Count × Weight
External Inputs (EI)	7	4	28
External Outputs (EO)	3	5	15
External Inquiries (EQ)	3	4	12
Internal Logical Files (ILF)	5	10	50
External Interface Files (EIF)	3	7	21
Unadjusted FP (UFP)			**126**

UFP = (7×4) + (3×5) + (3×4) + (5×10) + (3×7)
= 28 + 15 + 12 + 50 + 21 = **126**

Value Adjustment Factor (VAF) = 0.65 + (0.01 × Total Degree of Influence)

where DI = sum of ratings (0–5) across the 14 General System Characteristics (GSC).

No.	General System Characteristic	Rating (0–5)	Reason
1	Data communications	5	Constant interaction with external systems: payment gateway, maps/GPS, SMS/email services and restaurant apps.
2	Distributed data processing	4	Customer app, restaurant dashboard and delivery-partner app all process data on separate devices while sharing a central server.
3	Performance	4	Fast response needed during lunch/dinner peak hours for search, checkout and live order tracking.
4	Heavily used configuration	3	Many concurrent users, but runs on standard cloud infrastructure without special hardware limits.
5	Transaction rate	5	Very high number of orders, payments and status updates in peak hours and during discount offers.
6	Online data entry	5	Customers enter orders and addresses, restaurants update menus and stock, delivery partners update status continuously.
7	End-user efficiency	4	Simple UI needed: quick search, one-tap reorder, easy checkout and easy tracking.
8	Online update	5	Real-time update of order status, delivery partner GPS location and item availability.
9	Complex processing	3	Moderate: delivery-partner assignment, delivery fee/ETA calculation, coupon and discount rules.
10	Reusability	3	Modules such as login, payment and notification can be reused, but most logic is specific to food delivery.

No.	General System Characteristic	Rating (0–5)	Reason
11	Installation ease	2	Cloud-hosted web/mobile app; users only install the app, but restaurants need onboarding and API set-up.
12	Operational ease	3	Needs automated backups and monitoring; recovery must be fast but staff are trained technical users.
13	Multiple sites	4	Used in many cities, at many restaurants and by many delivery partners at the same time.
14	Facilitate change	4	Frequent changes to menus, prices, offers, delivery zones and commission rules must be easy to make.
	Total DI		**54**

Total DI Calculation:

DI = 5 + 4 + 4 + 3 + 5 + 5 + 4 + 5 + 3 + 3 + 2 + 3 + 4 + 4 = **54**
VAF = 0.65 + (0.01 × 54) = 0.65 + 0.54 = **1.19**

Final FP (after VAF):

FP = UFP × VAF = 126 × 1.19 = **149.94 FP ≈ 150 FP**

Language ratio: Java ≈ 53 LOC/FP

Estimated LOC:

LOC = 149.94 × 53 = **7,947 LOC** (≈ 7.95 KLOC)

Step 4: Convert Size to Effort, Time & Cost

A) Productivity Approach (simple)

- Assume productivity = 500 LOC / Person-Month (PM).
- Effort = LOC / Productivity

Effort = 7,947 / 500 = **15.89 PM**
Cost (salary ₹50,000 per PM) = 15.89 × 50,000 = ₹794,500 ≈ **₹7.95 Lakh**

B) Basic COCOMO (Semi-Detached – typical team and mixed experience)

Constants: **a_1 = 3.0, a_2 = 1.12, b_1 = 2.5, b_2 = 0.35**

- Size: KLOC = 7.947
- Effort (PM) = $a_1 \times (\text{KLOC})^{a2}$
- Time (months) = $b_1 \times (\text{Effort})^{b2}$
- Team size = Effort / Development time

Effort = $3.0 \times (7.947)^{1.12} = 3.0 \times 10.191 =$ **30.57 PM**
Time = $2.5 \times (30.57)^{0.35} = 2.5 \times 3.310 =$ **8.28 months**
Team size = 30.57 / 8.28 = 3.69 ≈ **4 persons** (3–4)

Cost: If salary per PM = ₹50,000

Total Cost = 30.57 × 50,000 = ₹1,528,500 ≈ **₹15.29 Lakh**

Comparison of the two approaches

Method	Effort (PM)	Time (months)	Team size	Cost (₹)
Productivity (500 LOC/PM)	15.89	–	–	794,500
Basic COCOMO (Semi-detached)	30.57	8.28	≈ 4	1,528,500

The simple productivity method gives a lower value because it assumes a fixed rate for every project. COCOMO is more realistic because effort grows faster than size (exponent 1.12), so it is taken as the final estimate.

Step 5: Present Results

"Estimated effort ≈ 31 PM, duration ≈ 8.3 months, team size ≈ 4 persons, and total cost ≈ ₹15.29 lakh for the Online Food Delivery System."

Conclusion

The size of the Online Food Delivery System was estimated using Function Points (150 FP) and converted to about 7,947 LOC in Java. Using Basic COCOMO (semi-detached mode), the project needs about 30.57 person-months of effort, about 8.28 months of development time, a team of about 4 persons and a budget of about ₹15.29 lakh. Thus the LOC, Function Point, Productivity and COCOMO techniques were studied and applied successfully.